

AD9082/AD9081/AD9988/AD9986 Python Example Application Release Notes

MPG SW System Apps

Exported on 10/22/2024

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1 AD9082/AD9081/AD9988/AD9986 Python Example Application Release Notes v2.4.0

1.1 Rev 2.4.0 Release on Oct 2024

1.2 Release Summary

This release will provide the follow key changes that are recommended:

- Integrate API v1.7.0 into Python example application.
- Updated NCO test

1.3 Deliverables Summary

Deliverable	Description	Release ID	Updated
fpga images	ADS9 Board Image		No
erpc_pkgs	PC-Evaluation Board Client-Server Includes Latest erpc client server files • .Net DLL AD9082/AD9081/AD9988/AD9986 client for Python Application • AD9082/AD9081/AD9988/AD9986 server for Microzed.	v1.7.0	Yes
data_control_libs	Data Capture and Data Playback Libraries .Net DLLs for Data Capture and Playback Libraries for AD9082/AD9081/AD9988/AD9986 Evaluation Platform		No
adxxxx_python_eval_app	Python Source Code Example App	v2.4.0	Yes

1.4 ADxxxx Python Eval App

1.4.1 New Features, Interface Changes:

Summary of adxxxx_python_eval_app changes.

Refer to python eval app source code for example usage.

Update	Reason for Update	API Updated	Required/ Recommended Optional	Usage
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1.4.2 Bug Fixes/ Improvements

Summary of bug fixes and improvements.

Bug Fixes/ Improvement	API Updated	Reason for Update	Required/ Recommended Optional
NCO test updated to use adi_adxxxx_device_startup_nco_test_mode	ConfigureNcoTestMode.configure()	Allows jesd mode to be passed into nco configuration	Optional

1.4.3 Known Issues

- Links down for UC 10 with external clocks.
- The python eval app currently does not configure the HMC7044 to provide a SYSREF signal for Subclass 1. Future releases will support these SYSREF configurations.

1.5 Regression Testing

Deliverable	Compilation Testing	System Testing (UCs 3,10,11,20)	Status
API Device Libraries	✓	✓	✓

1.6 HW Setup on Regression Testing

- FPGA platform: ADS9
- Evaluation Boards:
 - AD9081-FMCA-EBZ
 - AD9082-FMCA-EBZ
 - AD9986-FMCB-EBZ
 - AD9988-FMCB-EBZ

2 AD9082/AD9081/AD9988/AD9986 Python Example Application Release Notes v2.3.0

2.1 Rev 2.3.0 Release on Nov 2023

2.2 Release Summary

This release will provide the follow key changes that are recommended:

- Integrate API v1.6.0 into Python example application.

2.3 Deliverables Summary

Deliverable	Description	Release ID	Updated
fpga images	ADS9 Board Image		No
erpc_pkgs	PC-Evaluation Board Client-Server Includes Latest erpc client server files • .Net DLL AD9082/AD9081/AD9988/AD9986 client for Python Application • AD9082/AD9081/AD9988/AD9986 server for Microzed.	v1.6.0	Yes
data_control_libs	Data Capture and Data Playback Libraries .Net DLLs for Data Capture and Playback Libraries for AD9082/AD9081/AD9988/AD9986 Evaluation Platform		No
adxxxx_python_eval_app	Python Source Code Example App	v2.3.0	Yes

2.4 ADxxxx Python Eval App

2.4.1 New Features, Interface Changes:

Summary of adxxxx_python_eval_app changes.

Refer to python eval app source code for example usage.

Update	Reason for Update	API Updated	Required/Recommended Optional	Usage
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2.4.2 Bug Fixes/ Improvements

Summary of bug fixes and improvements.

Bug Fixes/ Improvement	API Updated	Reason for Update	Required/Recommended Optional
Print statement for number of captured words	real_data_test()	To keep track of number of captured samples	Optional

2.4.3 Known Issues

- Links down for UC 10 with external clocks.
- The python eval app currently does not configure the HMC7044 to provide a SYSREF signal for Subclass 1. Future releases will support these SYSREF configurations.

2.5 Regression Testing

Deliverable	Compilation Testing	System Testing (UCs 3,10,11,20)	Status
API Device Libraries	✓	✓	✓

2.6 HW Setup on Regression Testing

- FPGA platform: ADS9
- Evaluation Boards:
 - AD9081-FMCA-EBZ
 - AD9082-FMCA-EBZ
 - AD9986-FMCA-EBZ
 - AD9988-FMCA-EBZ

3 AD9082/AD9081/AD9988/AD9986 Python Example Application Release Notes v2.2.0

3.1 Rev 2.2.0 Release on June 2023

3.2 Release Summary

This release will provide the follow key changes that are recommended:

- Integrate API v1.5.0 into Python example application.
- Support M=1 use cases

3.3 Deliverables Summary

Deliverable	Description	Release ID	Updated
fpga images	ADS9 Board Image		No
erpc_pkgs	PC-Evaluation Board Client-Server Includes Latest erpc client server files • .Net DLL AD9082/AD9081/AD9988/AD9986 client for Python Application • AD9082/AD9081/AD9988/AD9986 server for Microzed.	v1.5.0	Yes
data_control_libs	Data Capture and Data Playback Libraries .Net DLLs for Data Capture and Playback Libraries for AD9082/AD9081/AD9988/AD9986 Evaluation Platform		No
adxxxx_python_eval_app	Python Source Code Example App	v2.2.0	Yes

3.4 ADxxxx Python Eval App

3.4.1 New Features, Interface Changes:

Summary of adxxxx_python_eval_app changes.

Refer to python eval app source code for example usage.

Update	Reason for Update	API Updated	Required/Recommended Optional	Usage
Added command line input "--realonly" 1/0	To support real data	N/A	Optional	Cmd line input: python txfe_main.py --uc UC --realonly 1
FPGA register sequence for jesd configuration	Improve stability	ConfigureFullChipMode.configure()	Recommended	configure(uc, 1, "ce")
Add real_data_test() in main app	To support M=1 use cases	New API: real_data_test()	Optional	real_data_test()

3.4.2 Bug Fixes/ Improvements

Summary of bug fixes and improvements.

Bug Fixes/Improvement	API Updated	Reason for Update	Required/Recommended Optional
Timeout for fifo_read_ready register	txfe_dpg_wrapper.py capture()	Prevent infinite readback	Recommended

Bug Fixes/ Improvement	API Updated	Reason for Update	Required/Recommended Optional
Support M=1 when generating vectors	create_iq_vecs()	Support real mode UCs	Optional

3.4.3 Known Issues

- Links down for UC 10 with external clocks.
- The python eval app currently does not configure the HMC7044 to provide a SYSREF signal for Subclass 1. Future releases will support these SYSREF configurations.

3.5 Regression Testing

Deliverable	Compilation Testing	System Testing (UCs 3,10,11,20)	Status
API Device Libraries	✓	✓	✓

3.6 HW Setup on Regression Testing

- FPGA platform: ADS9
- Evaluation Boards:
 - AD9081-FMCA-EBZ
 - AD9082-FMCA-EBZ
 - AD9986-FMCB-EBZ
 - AD9988-FMCB-EBZ

4 AD9082/AD9081/AD9988/AD9986 Python Example Application Release Notes v2.1.0

4.1 Rev 2.1.0 Release on Dec 2022

4.2 Release Summary

This release will provide the follow key changes that are recommended:

- Integrate API v1.4.0 into Python example application.

4.3 Deliverables Summary

Deliverable	Description	Release ID	Updated
fpga images	ADS9 Board Image		No
erpc_pkgs	PC-Evaluation Board Client-Server Includes Latest erpc client server files • .Net DLL AD9082/AD9081/AD9988/AD9986 client for Python Application • AD9082/AD9081/AD9988/AD9986 server for Microzed.	v1.4.0	Yes
data_control_libs	Data Capture and Data Playback Libraries .Net DLLs for Data Capture and Playback Libraries for AD9082/AD9081/AD9988/AD9986 Evaluation Platform		No
adxxxx_python_eval_app	Python Source Code Example App	v2.1.0	Yes

4.4 ADxxxx Python Eval App

4.4.1 New Features, Interface Changes:

Summary of adxxxx_python_eval_app changes.

Refer to python eval app source code for example usage.

Update	Reason for Update	API Updated	Required/Recommended Optional	Usage
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4.4.2 Bug Fixes/ Improvements

Summary of bug fixes and improvements.

Bug Fixes/Improvement	API Updated	Reason for Update	Required/Recommended Optional
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4.4.3 Known Issues

- Links down for UC 10 with external clocks.
- The python eval app currently does not configure the HMC7044 to provide a SYSREF signal for Subclass 1. Future releases will support these SYSREF configurations.

4.5 Regression Testing

Deliverable	Compilation Testing	System Testing (UCs 3,10,11,20)	Status
API Device Libraries	✓	✓	✓

4.6 HW Setup on Regression Testing

- FPGA platform: ADS9

- Evaluation Boards:
 - AD9081-FMCA-EBZ
 - AD9082-FMCA-EBZ
 - AD9986-FMCB-EBZ
 - AD9988-FMCB-EBZ

5 AD9082/AD9081/AD9988/AD9986 Python Example Application Release Notes v2.0.0

5.1 Rev 2.0.0 Release on July 2022

5.2 Release Summary

This release will provide the follow key changes that are recommended:

- Integrate API v1.3.1 into Python example application.
- Included fpga image that supports Subclass 1 on ADS9 board.
- Updated the Pipfile to specify versioning for external packages.

5.3 Deliverables Summary

Deliverable	Description	Release ID	Updated
fpga images	ADS9 Board Image		Yes
erpc_pkgs	PC-Evaluation Board Client-Server Includes Latest erpc client server files • .Net DLL AD9082/AD9081/AD9988/AD9986 client for Python Application • AD9082/AD9081/AD9988/AD9986 server for Microzed.	v1.3.1	Yes
data_control_libs	Data Capture and Data Playback Libraries .Net DLLs for Data Capture and Playback Libraries for AD9082/AD9081/AD9988/AD9986 Evaluation Platform		No
adxxxx_python_eval_app	Python Source Code Example App	v2.0.0	Yes

5.4 ADxxxx Python Eval App

5.4.1 New Features, Interface Changes:

Summary of adxxxx_python_eval_app changes.

Refer to python eval app source code for example usage.

Update	Reason for Update	API Updated	Required/Recommended Optional	Usage
Added versioning to the Pipfile	Standardize external package versioning	N/A	Optional	During installation

5.4.2 Bug Fixes/ Improvements

Summary of bug fixes and improvements.

Bug Fixes/Improvement	API Updated	Reason for Update	Required/Recommended Optional
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5.4.3 Known Issues

- The python eval app does not support ADC full BW mode at this time.
- Links down for UC 10 with external clocks.
- The python eval app currently does not configure the HMC7044 to provide a SYSREF signal for Subclass 1. Future releases will support these SYSREF configurations.

5.5 Regression Testing

Deliverable	Compilation Testing	System Testing (UCs 3,10,11,20)	Status
API Device Libraries	✓	✓	✓

5.6 HW Setup on Regression Testing

- FPGA platform: ADS9
- Evaluation Boards:
 - AD9081-FMCA-EBZ
 - AD9082-FMCA-EBZ
 - AD9986-FMCB-EBZ
 - AD9988-FMCB-EBZ

6 AD9082/AD9081/AD9988/AD9986 Python Example Application Release Notes v1.2.4

6.1 Rev 1.2.4 Release on March 2022

6.2 Release Summary

This release will provide the follow key fixes that are recommended:

- Added oneshot sync API to application level

6.3 Deliverables Summary

Deliverable	Description	Release ID	Updated
fpga images	ADS9 Board Image		No
erpc_pkgs	PC-Evaluation Board Client-Server Includes Latest erpc client server files • .Net DLL AD9082/AD9081/AD9988/AD9986 client for Python Application • AD9082/AD9081/AD9988/AD9986 server for Microzed.	v1.2.2	Yes
data_control_libraries	Data Capture and Data Playback Libraries .Net DLLs for Data Capture and Playback Libraries for AD9082/AD9081/AD9988/AD9986 Evaluation Platform		No
adxxxx_python_eval_app	Python Source Code Example App	v1.2.4	Yes

6.4 ADxxxx Python Eval App

6.4.1 New Features, Interface Changes:

Summary of adxxxx_python_eval_app changes.

Refer to python eval app source code for example usage.

Update	Reason for Update	API Updated	Required/Recommended Optional	Usage
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6.4.2 Bug Fixes/ Improvements

Summary of bug fixes and improvements.

Bug Fixes/ Improvement	API Updated	Reason for Update	Required/Recommended Optional
Implemented oneshot sync on app level	configure_full_chip_mode.py	Synchronization wasn't occurring causing links down for certain use cases	Recommended

6.4.3 Known Issues

- The python eval app does not support ADC full BW mode at this time.
- Links down for UC 10 with external clocks

6.5 Regression Testing

Deliverable	Compilation Testing	System Testing (UCs 3,10,11,20)	Status
API Device Libraries	✓	✓	✓

6.6 HW Setup on Regression Testing

- FPGA platform: ADS9
- Evaluation Boards:
 - AD9081-FMCA-EBZ
 - AD9082-FMCA-EBZ
 - AD9986-FMCA-EBZ
 - AD9988-FMCA-EBZ

7 AD9082/AD9081/AD9988/AD9986 Python Example Application Release Notes v1.2.3

7.1 Rev 1.2.3 Release on March 2022

7.2 Release Summary

This release will provide the follow key fixes that are recommended:

- Update to auto configure device
- Remove duplicate vector download and playback

7.3 Deliverables Summary

Deliverable	Description	Release ID	Updated
fpga images	ADS9 Board Image		No
erpc_pkgs	PC-Evaluation Board Client-Server Includes Latest erpc client server files • .Net DLL AD9082/AD9081/AD9988/AD9986 client for Python Application • AD9082/AD9081/AD9988/AD9986 server for Microzed.	v1.2.2	Yes
data_control_libraries	Data Capture and Data Playback Libraries .Net DLLs for Data Capture and Playback Libraries for AD9082/AD9081/AD9988/AD9986 Evaluation Platform		No
adxxxx_python_eval_app	Python Source Code Example App	v1.2.3	Yes

7.4 ADxxxx Python Eval App

7.4.1 New Features, Interface Changes:

Summary of adxxxx_python_eval_app changes.

Refer to python eval app source code for example usage.

Update	Reason for Update	API Updated	Required/Recommended Optional	Usage
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7.4.2 Bug Fixes/ Improvements

Summary of bug fixes and improvements.

Bug Fixes/ Improvement	API Updated	Reason for Update	Required/Recommended Optional
Auto configures "device" variable	settings.py erpc_factory.py ad9082_client.py eval_board_target.py	Needed to manually edit the device variable before running program	Recommended
Removed duplicate vector download and playback	txfe_main.py configure_full_chip_mode.py	Unnecessary duplicate, slows down main loopback function	Recommended

7.4.3 Known Issues

- The python eval app does not support ADC full BW mode at this time.

7.5 Regression Testing

Deliverable	Compilation Testing	System Testing (UCs 3,10,11,20)	Status
API Device Libraries	✓	✓	✓

7.6 HW Setup on Regression Testing

- FPGA platform: ADS9
- Evaluation Boards:
 - AD9081-FMCA-EBZ
 - AD9082-FMCA-EBZ
 - AD9986-FMCB-EBZ
 - AD9988-FMCB-EBZ

8 AD9082/AD9081/AD9988/AD9986 Python Example Application Release Notes v1.2.2

8.1 Rev 1.2.2 Release on Feb 2022

8.2 Release Summary

This release will provide the follow key fixes that are recommended:

- Initial public release
- Accompanies eRPC Package v1.2.2 of the API
- Update to settings.py folder to choose device
- Update to capture side sample manipulation

8.3 Deliverables Summary

Deliverable	Description	Release ID	Updated
fpga images	ADS9 Board Image		No
erpc_pkgs	PC-Evaluation Board Client-Server Includes Latest erpc client server files • .Net DLL AD9082/AD9081/AD9988/AD9986 client for Python Application • AD9082/AD9081/AD9988/AD9986 server for Microzed.	v1.2.2	Yes
data_control_libraries	Data Capture and Data Playback Libraries .Net DLLs for Data Capture and Playback Libraries for AD9082/AD9081/AD9988/AD9986 Evaluation Platform		No
adxxx_python_eval_app	Python Source Code Example App	v1.2.2	Yes

8.4 ADxxxx Python Eval App

8.4.1 New Features, Interface Changes:

Summary of adxxxx_python_eval_app changes.

Refer to python eval app source code for example usage.

Update	Reason for Update	API Updated	Required/Recommended Optional	Usage
Added "device" variable in settings for user to choose hardware specifications	Ability to use exported erpc package associated with device	adxxxx_python_eval_app/ settings.py variable "device"	Recommended	settings.py
Added differing jesd parameter duallink capability in capture.	Capture samples assumed jesd parameters to be equal in duallink case	adxxxx_python_eval_app/ capture_deframer.py def get_samples(self, input32, num_links, jesd_m_link0, jesd_m_link1=0)	Recommended	txfe_main.py

8.4.2 Bug Fixes/ Improvements

Summary of bug fixes and improvements.

Bug Fixes/Improvement	API Updated	Reason for Update	Required/Recommended Optional
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8.4.3 Known Issues

- The python eval app does not support ADC full BW mode at this time.

8.5 Regression Testing

Deliverable	Compilation Testing	System Testing (UCs 3,10,11,20)	Status
API Device Libraries	✓	✓	✓

8.6 HW Setup on Regression Testing

- FPGA platform: ADS9
- Evaluation Boards:
 - AD9081-FMCA-EBZ
 - AD9082-FMCA-EBZ
 - AD9986-FMCB-EBZ
 - AD9988-FMCB-EBZ

9 AD9082/AD9081/AD9988/AD9986 Python Example Application Release Notes v1.2.1 and below

9.1 Initial development